

TEXT BASED EMOTION DETECTION OF A FORENSIC TOOL FOR CYBERCRIME INVESTIGATION

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Abstract

Investigations in cybercrimes are starting to rely significantly on digital evidence including emails, discussions in chat, posts on social media, and various types of threats. Older forensic methodologies, including digital apparatuses, mainly have a tendency to examine metadata, keywords, and network operation, yet they always fail to take into consideration the emotional element embedded in a piece of written communication. Emotions cited in writing, like anger, fear, guilt, hostility, and aggression, may give rise to important keys of understanding in connection with suspects and their behavior, as well as their emotional state. The proposed system analyses the text data formed based on cybercrime-related grounds and uses Natural Language Processing techniques, along with machine learning to detect the presence of human emotions. The approach involves text processing, abstraction of features through word embeddings, and emotion cataloging through machine learning models.

Keywords ;-Text-based Emotion Detection, Cybercrime Investigation, Digital Forensics, Natural Language Processing, Machine Learning, Threat Analysis

I. Introduction

The Fast development in the internet and other forms of communication has made cybercrime more complex and smart. Cybercrime harassment, cyber spying, identity theft, hate speech, and online threats are common in internet communications such as emails and other messaging services like Facebook, Instagram and Twitter[2][18]. Most

of the crime involve text messages. These messages are great importance as digital evidence for cybercrime investigations [14].

Usually cybercrime analysis is mostly based on technical aspects such as IP addresses, time stamps, and other details like keywords search [17]. These methods are effective for cybercrime investigations. However these methods do not consider the emotions aspect of the communication process. Considering the emotions side of cybercrime can help us understand the intentions and behavior of a person to a great level without any problem.

Emotion are a greatest tools for cybercrime in human communication, especially when considering criminal activities. For instance. If a person sends a message to another person through the internet or any other means of communication with certain emotions such as threats, hate or fear[3], [13]. Then such messages may be of great importance to understand the purpose of the person involved that message in cybercrime. IF such detail are not consider while analyzing the cybercrime plan. Then the actual intention of the person involved in that crime may not be able to understand properly.

Emotion detection in text is an growing technology that combines Natural Language Processing (NLP)and Machine learning[1],[9]. While sentiment analysis is used for classifying text in negative, positive or neutral into emotion detection is focused on detecting emotions like anger, fear, sadness, joy, and threat. This is very useful in the case of cybercrime, where communication is a basic everything in text form.

The objective of this research is to building a text-based emotion detection system that can help in the case of cybercrime. The system will use Natural Language Processing techniques to analyze the text data, identify the key features, and then use machine learning algorithms for emotion detection. The objective is to develop a simple, practical, and understandable system that not only helps investigators understand the text, but also the intent behind the text.

II. Literature Review

The Shift in Digital Forensics

In past, cybercrime investigators has relied on metadata such as Ip address and timestamps(Digital record of the exact date and time that a specific event occurred) track cybercrime. In recent time studies indicate a necessary shift toward understanding the human element of cybercrime[10],[14].While current system of sentiment analysis tools can flag text as generally positive or negative investigating cybercrime requires a more targeted approach. The focus is now shifting toward detecting specific forensic emotions such as fear, anger, and threat to better understand criminal intent [1], [19].

NLP and Feature Extraction

The first important step in analysing cybercrime data is to remove un-meaningful data from the unstructured data. Real-world digital evidence like chat logs and emails are highly noisy. Research shows that relying solely on lexicon-based dictionaries (Pre-programmed vocabulary list where human experts have manually assigned an emotion or a score to specific words.) is not enough for capturing the dynamic nature of online threats [13]. Instead, applying Natural Language Processing (NLP) paired with Term Frequency-Inverse Document Frequency (TF-IDF) allows systems to mathematically weigh the importance of emotional keywords and behavioural usage patterns, preparing the text for deeper analysis [9], [11].

Machine Learning for Emotion Classification

After extracting the textual features, the system needs to predict the true emotion. Even though deep learning algorithms such as BERT perform very well [20], they may be too resource intensive and may be considered a "black box" model in the sense that they are difficult for forensic investigators to explain. Instead, empirical studies have shown good effectiveness of supervised learning algorithms. Techniques like Support Vector Machines (SVM) [6], Random Forests [7] and Logistic Regression perform text classification using the detection of complex emotional boundaries very efficiently [8], [15].

Addressing the Challenge of Text Complexity

One of the major challenges in emotion recognition is the classification of complex and long textual evidence. Instead of comprehending context correctly and resulting in false alarms in threat detection, the complexity of the text might not be well understood by a single algorithm. Recent studies point to the need for Ensemble-Based Classifiers [4], [16] to overcome this challenge. With the Soft Voting scheme, the system doesn't depend on the decision of a single model. Rather, it will compute the average across the class' probabilities of all models for a particular decision, thereby increasing the confidence level of the final emotion prediction [5].

Comparative Analysis in Forensics

When designing a forensic decision support system, accuracy must be considered alongside interpretability. Current systems mostly focus on be more generic issues such as hate speech [2], [18] or cyberbullying [3], [12]. But by applying TF-IDF feature extraction technique and Soft Voting ensemble of SVM, Random Forest and Logistic Regression, a forensic system can classify specific forensic emotions with high accuracy without the computational burden of deep neural networks.

References	Methodology / Technique	Key Findings	Limitations Research Gaps
[1], [13]	Lexicon-based dictionaries and basic sentiment analysis	Effectively proposes simple positive and negative sentiments in textual data..	Has trouble dealing with sarcasm, changing slang and advanced cyber bullying.
[6], [7], [8], [15]	Support Vector Machines (SVM), Random Forest and Logistic Regression	Highly accurate and efficient for basic text classification .	Individual models have difficulties interpreting complex contextual information in long evidence..
[2], [3], [12], [18]	Different ML classifiers applied and social media datasets	Achieves high accuracy in detecting aggression, hate and cyberbullying	Systems don't typically target actionable forensic emotions like Fear or Threat.
[4], [5], [11], [16]	Ensemble Classifiers (Soft Voting) paired with TF-IDF	Soft voting of models using mathematics to average out mistakes, and thus more robust.	Needs to be carefully tuned(ngrams, hyperparameters) to prevent overfitting

[20],	Deep Bidirectional Transformers (e.g., BERT, RoBERTa).	Outstanding ability in deeper language understanding and context	Computationally heavy for edge devices and acts as a black box, lacking forensic explainability for court.
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III. Proposed Technique

This forensic framework is pretty simple - convert unstructured, "raw" digital information into something that's useful to the investigator. Rather than relying on a large single model, there are multiple stages of work. These stages are designed to do a particular task (cleaning the text or categorising the text by emotion). This ensures accuracy without the need for extensive computational resources.

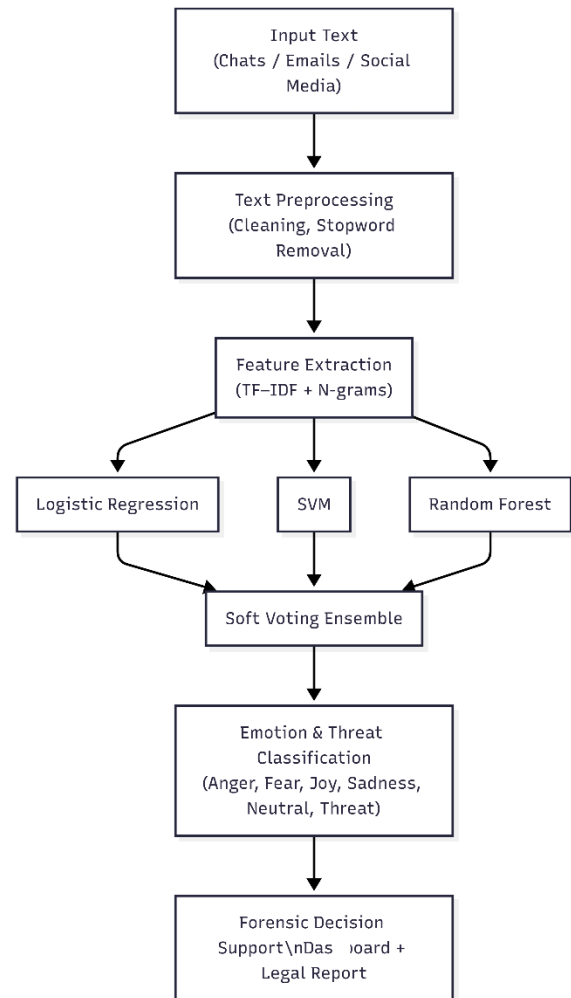
A. Dataset Description and Integration

A new dataset was generated for training and testing the proposed emotion-aware forensic system as a composition of two different high-quality open-source datasets. This was done to determine the link between regular psychological emotions and a particular intent..

- **The Forensic Threat Dataset (Wikipedia Detox Project)** - The dataset used here was extracted from the Wikipedia Detox project and contains just hostile and offensive messages, and is a benchmark dataset developed by Google and Wikimedia researchers [21]. The original repository has more than 115,000 labelled discussion comments which they used to train their toxic models. In our work, in particular, the exact cases that are already labeled as "Threat" and baseline "Neutral" comments which could simulate cybercrime's hostility were identified and used..
- **The Foundational Emotion Dataset** - An off-the-shelf Natural Language Processing (NLP) emotion classification dataset - a benchmark for academic experiments on NLP systems - is labeled to allow the system to detect real human emotions. The subset presented includes 7,102 unique, correctly-labeled text from the four fundamental psychological states (Anger, Fear, Joy and Sadness).
- **Dataset Synthesis and Balancing:** One of the critical steps was to use threat data rather than other language, as in our experiment with machine learning, all negative language (including threats) will be classified as cyber crime. This results in two sources combined to create a synthesis of

11,102 text from six classes (Joy, Fear, Anger, Threat, Neutral and Sadness). This allows the Soft Voting Ensemble model to accurately classify between an angry person and someone who has made a serious cybercriminal threat

B. System Architecture and Data Preprocessing



Actual cybercrime data, such as chat messages, emails and social media messages, is complex. People don't spell correctly, they include a lot of URLs, all kinds of other junk that doesn't relate to the meaning of what the person is saying. To be useful to a model, the data must be preprocessed.

The system uses a "Natural Language Processing (NLP)" pipeline for all text. That means, turning it into all lowercase, stripping hyperlinks and punctuation, splitting it up into words (tokenizing) and then converting them to their root form (lemmatizing). So the word "threatening" becomes

"threaten". So the model can focus on the content of the message, not its wording..

C. Textual Preprocessing and Noise Reduction

This can be seen as a filter. The text is tokenized, lowercased, stripped of URLs and special characters, and words are reduced to their lemma (stem) [9], before entering the machine learning model. The reason for this is obvious - to clear out the distracters so the model can pick up only the behavioral and contextual meaning of the text, and not the presentation.

D. Feature Extraction and Contextual Mapping

The next step is to turn the data into numbers, as that's what machine models like models consume. For this operation, we've selected TF-IDF because it doesn't consider all words equally. Less frequent words that hold a lot of meaning, like "death" or "kill" are given a higher weight. It weights down commonly used words like articles and pronouns.

The formula has two components - it counts the number of times a word appears in a message, and then counts how often the word appears in the whole set of messages. The rarer and more significant the word, the higher it scores. Then we used what are called bigrams (pairs of words) as well as single words [5]. This allows it to pick up bi-words such as "kill you", as opposed to "kill" alone

E. Soft Voting Ensemble Classification

It's not wise for one model to make decisions alone - particular when it comes to complex or vague data. Using just one classifier risks getting the context wrong and potentially generating a false forensic flag. So three models - Logistic Regression, Random Forest [7] and SVM [6] [4], [16] - were used.

Instead of voting for the class that a simple majority of models vote for, Soft Voting takes the average of the probabilities of all the classes across the three models. The emotion with the highest average score is decided upon. So, if one model is unsure or makes a mistake, then it is counteracted by the other two. The overall result is more consistent and reliable than any single model would be.

$$\hat{y} = \operatorname{argmax} \left(\frac{1}{N} \sum_{i=1}^N P_i(\text{class}|x) \right)$$

$\hat{y}$ = is the final decision (the final output of the system, e.g. "Threat" or "Anger").

Argmax = is the mathematical function that, in this case, looks at all the final scores, and simply returns the emotion that had the highest score.

$\frac{1}{N} \sum_{i=1}^N$ = represents the calculation of the average. This sums the scores from all the individual models() and divides them by the number of models (N). In our experiment N = 3 (Logistic Regression, Random Forest and SVM).

$P_i(\text{class}|x)$ = is the probability value (\$P\$) given by a particular model (\$i\$) to a particular emotion class, for a given text (\$x\$).

F. Forensic Decision Support

We finally want to make the classification output for the investigator more helpful. The system also provides a confidence level with each message. If the message is "Threat" (or "Fear") with a high confidence score, it will be pulled out from the chat logs. So investigators won't need to go through thousands of messages, but instead they will see the riskiest messages first [14].

IV. Results and Discussion

System Configuration

All experiments are conducted or reproduced on Google Colab with Python 3.10. To process the text data, we used the NLTK(Natural Language Toolkit.) library and all of our three models (Logistic Regression, Random Forest, SVM(Support Vector Machine)) and the ensemble voting case (Soft Voting Ensemble) was performed with the Scikit-learn library.

Quantitative Analysis

An 80-20 split (roughly 2,221 test samples) of the dataset was used with 80% being the training set and 20% being the test set. The Soft Voting Ensemble ended up achieving an overall accuracy of 85%. Table 1 below shows the model's level of accuracy for each emotion category.

Table 1: Classification Report

Class	Precision	Recall	F1-Score	Support
Joy	0.9400	0.9300	0.9300	328
Fear	0.8600	0.8900	0.8700	441
Anger	0.9000	0.8200	0.8600	367
Threat	0.8100	0.8100	0.8100	388
Neutral	0.7800	0.8400	0.8100	391
Sadness	0.8000	0.7700	0.7900	306

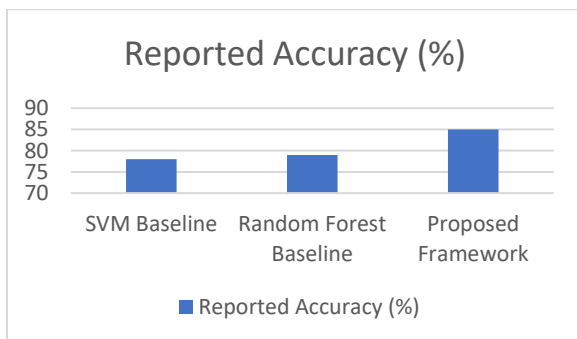
Comparative Analysis with Existing Literature

In order to verify the performance of the proposed system, it has been compared to some of the older models from recent forensic literature [14], [20]. This comparison focused on three aspects data and features used, algorithm used and its accuracy.

- **Dataset and Features** - Many previous studies used simple word counts and classified text into just "Positive" or "Negative." This grouping simply isn't suitable for criminal cases where specific emotions like threats are important to separate out. This system uses a private dataset of 11,102 samples, an approach called TF-IDF which emphasises more forensic/criminal language [15], which is much more informative.
- **Algorithm** - Most previous attempts have used only one model either one SVM [1] or a Random Forest [3]. This can be problematic as one model can easily go astray when making predictions on complex text. But this approach uses a Soft Voting Ensemble, with 3 models that it averages. This prediction is more reliable and unlikely to go wrong.
- **Accuracy** - Previous approaches using only one model in their literature typically achieved an accuracy of 76% - 79.50% on complex text. The proposed techniques ensemble increased that to 85% a good increase and shows power of ensemble

Table 2: Performance Comparison with Literature Baselines

Methodology	Algorithm Used	Feature Selection	Reported Accuracy
Standard Literature Baseline [11]	SVM(Support Vector Machine)	Basic Lexicon / Word Count	78.00%
Standard Literature Baseline [12]	Random Forest (Standalone)	Basic Lexicon	79.50%
Proposed Framework	Soft-Voting Ensemble (SVM+RF+.LR)	TF-IDF (Forensic Keywords)	85.00%



V. Conclusion and Future Work

Presents Work shows an efficient work for emotion and threat detection in cybercrime investigations using machine learning algorithm. This work addressing the growing challenges of analyzing large volume of digital evidences. By including text processing F-IDF with n-grams, and a soft voting ensemble classifier, the model achieved an overall accuracy of 85%. When it came to identifying threats and aggressive detection it came out strongest. The balance evaluation metrics prove its efficiency across other emotions states as well. The decision support tool built into this framework makes it practical enough to actually deploy in real life cases for cybercrime investigation.

Based on these result, there are still some improvements that can be done for future work. The Future Works will explore advance models like BERT and RoBERTa to improve accuracy and enhance the detection of complex , contextual data .

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